

THOHOYANDOU, South Africa

WMO: 681830

Lat: 23.0797S

Lon: 30.3836E

Elev: 2014

StdP: 13.66

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.3	46.5	24.5	20.0	70.9	28.6	24.3	71.3	15.4	68.6	13.6	68.3	7.3	260	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.0	96.7	67.0	94.0	67.2	91.5	67.6	74.9	85.0	73.7	83.7	72.7	82.6	7.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.1	128.2	78.9	71.0	123.0	77.9	70.0	118.9	77.1	39.9	85.7	38.8	83.6	37.8	82.8	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.3	11.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			37.4	76.8	2.8	1.4	35.4	77.8	33.8	78.6	32.2	79.4	30.2	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.4	76.1	76.5	74.5	69.6	66.2	62.2	61.3	65.4	69.9	72.6	74.8	75.6
	DBStd	7.19	4.37	3.96	4.05	4.13	3.99	4.23	4.23	4.82	5.50	5.98	6.25	6.42
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	373	1	0	1	10	31	103	129	54	18	10	8	8
	CDD50	7435	810	743	761	587	504	367	351	477	597	700	744	794
	CDD65	2333	346	323	297	148	70	20	15	66	165	244	302	337
	CDH74	21540	2730	2534	2181	1060	762	372	359	966	1992	2524	2961	3099
	CDH80	8557	1002	948	786	322	191	66	73	320	899	1184	1362	1404
Wind	WSAvg	5.5	5.5	5.2	4.9	4.8	5.3	5.9	5.5	5.8	6.1	6.2	5.8	5.4
Precipitation	PrecAvg	42.0	8.8	8.0	5.7	2.0	0.8	0.6	0.7	0.6	1.0	2.2	4.5	6.3
	PrecMax	102.0	26.4	47.9	18.4	6.9	4.2	2.6	4.2	2.0	5.0	6.4	10.8	13.1
	PrecMin	23.6	1.2	1.1	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	1.1	2.8
	PrecStd	17.0	5.7	9.3	4.3	1.6	0.9	0.7	0.9	0.5	1.3	1.7	2.1	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.9	96.4	95.4	90.3	89.7	85.3	85.7	90.9	97.9	100.0	98.7	98.3
		MCWB	69.5	69.9	69.6	66.4	63.2	59.1	56.2	61.5	62.1	64.6	66.3	70.7
	2%	DB	91.3	91.6	90.8	87.2	85.0	81.7	81.9	87.2	93.2	95.3	95.3	94.7
		MCWB	70.7	71.0	69.9	67.1	62.5	58.0	56.1	59.9	61.6	64.9	66.8	69.8
	5%	DB	88.7	88.8	88.0	83.9	82.0	78.6	78.1	83.9	89.5	91.5	91.8	91.4
		MCWB	70.2	70.7	70.1	67.0	61.8	57.9	55.6	58.4	61.2	64.8	67.6	69.9
	10%	DB	85.8	86.2	84.8	80.6	78.6	75.0	73.9	79.7	85.6	86.9	88.2	88.0
		MCWB	70.5	70.4	70.1	66.2	61.6	57.5	55.9	58.2	60.9	64.7	67.7	69.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	77.7	75.4	72.1	69.5	66.2	61.5	64.8	67.3	71.0	73.8	75.4
		MCDB	87.6	85.6	85.2	82.8	80.3	75.4	71.3	80.9	83.1	87.3	84.6	86.4
	2%	WB	74.4	75.0	73.7	70.4	66.7	62.9	59.8	63.0	65.6	69.3	72.6	73.7
		MCDB	85.3	83.3	83.4	79.7	77.4	72.1	70.8	78.1	81.1	84.9	83.7	84.7
	5%	WB	73.1	73.4	72.5	69.2	64.9	61.0	58.6	61.7	64.4	67.9	71.3	72.4
		MCDB	82.3	82.9	82.1	78.0	75.4	70.7	69.5	75.9	80.1	82.4	83.1	83.9
	10%	WB	72.0	72.0	71.4	68.0	63.3	59.3	57.5	60.3	63.0	66.5	70.0	71.3
		MCDB	80.8	81.2	80.6	76.9	73.6	69.3	68.9	73.5	77.8	80.0	80.7	83.0
Mean Daily Temperature Range	5% DB	MDBR	16.2	17.0	17.3	19.1	23.8	25.2	23.7	24.9	25.1	21.7	19.8	18.0
		MCDBR	22.2	22.4	24.2	26.8	29.5	30.7	31.7	33.1	34.1	31.3	27.8	25.6
		MCWBR	6.9	7.0	8.2	10.6	11.6	12.8	12.1	11.7	11.6	9.7	8.0	7.3
	5% WB	MCDBR	17.8	17.3	18.6	19.6	23.3	23.3	23.1	25.9	24.9	23.2	20.9	20.1
		MCWBR	6.9	7.2	7.7	8.7	11.1	11.7	11.2	10.3	10.1	8.3	7.9	6.9
Clear-Sky Solar Irradiance	taub	0.401	0.396	0.386	0.369	0.335	0.322	0.325	0.376	0.444	0.453	0.419	0.412	
	taud	2.390	2.403	2.424	2.426	2.466	2.481	2.467	2.310	2.154	2.179	2.311	2.356	
	Ebn at Noon	299	297	291	282	279	277	280	275	269	277	292	297	
	Edh at Noon	41	40	37	34	30	28	30	38	48	49	44	43	
All-Sky Solar Radiation	RadAvg	1949	1894	1702	1469	1381	1226	1289	1525	1722	1820	1891	1915	
	RadStd	153	188	147	101	68	90	61	83	120	123	164	188	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page